

Table 5(a-c): Present Weather Element and Measurement Codes

Weather codes (AU / AW / MW) describe precipitation or obstructions to vision occurring at the time of observation. It is not uncommon for one type of element to be reported without another. In other words, it is possible to have an AU element without an AW element or MW element. Depending on equipment used at the location, some automated stations report AW codes either with or instead of AU codes. Manually augmented stations report MW codes either with or instead of AU and AW codes. Definitions of element and measurement codes:

Table 5(a) – pres_wx_MW1[2][3] (sourced from manual reports)

- 00-Cloud development not observed or not observable
- 01-Clouds generally dissolving or becoming less developed
- 02-State of sky on the whole unchanged
- 03-Clouds generally forming or developing
- FU:04-Visibility-reduced-by-smoke,-e.g.-veldt-or-forest-fires,-industrial-smoke-or-volcanic-ashes
- HZ:05-Haze
- DU:06-Widespread-dust-in-suspension-in-the-air,-not-raised-by-wind-at-or-near-the-station-at-the-time-of-observation
- DU:07-Dust-or-sand-raised-by-wind-at-or-near-the-station-at-the-time-of-observation-but-no-well-developed-dust-whirl(s)-or-sand-whirl(s),-and-no-duststorm-or-sandstorm-seen-or,-in-the-case-of-ships,-blowing-spray-at-the-station
- DU:08-Well-developed-dust-whirl(s)-or-sand-whirl(s)-seen-at-or-near-the-station-during-the-preceding-hour-or-at-the-time-of-observation,-but-no-duststorm-or-sandstorm
- DU:09-Duststorm-or-sandstorm-within-sight-at-the-time-of-observation,-or-at-the-station-during-the-preceding-hour
- FG:11-Patches-of-shallow-fog-or-ice-fog-at-the-station,-whether-on-land-or-sea,-not-deeper-than-about-2-meters-on-land-or-10-meters-at-sea
- FG:12-More-or-less-continuous-shallow-fog-or-ice-fog-at-the-station,-whether-on-land-or-sea,-not-deeper-than-about-2-meters-on-land-or-10-meters-at-sea
- 13-Lightning-visible,-no-thunder-heard
- 14-Precipitation-within-sight,-not-reaching-the-ground-or-the-surface-of-the-sea
- 15-Precipitation-within-sight,-reaching-the-ground-or-the-surface-of-the-sea,-but-distant,-i.e.,-estimated-to-be-more-than-5-km-from-the-station
- 16-Precipitation-within-sight,-reaching-the-ground-or-the-surface-of-the-sea,-near-to,-but-not-at-the-station
- TS:17-Thunderstorm,-but-no-precipitation-at-the-time-of-observation
- SQ:18-Squalls-at-or-within-sight-of-the-station-during-the-preceding-hour-or-at-the-time-of-observation
- FC:19-Funnel-cloud(s)-(Tornado-cloud-or-waterspout)-at-or-within-sight-of-the-station-during-the-preceding-hour-or-at-the-time-of-observation
- 20-Drizzle-(not-freezing)-or-snow-grains-not-falling-as-shower(s)-(during-the-preceding-hour-but-not-at-the-time-of-observation)
- 21-Rain-(not-freezing)-not-falling-as-shower(s)-(during-the-preceding-hour-but-not-at-the-time-of-observation))
- 22-Snow-not-falling-as-shower(s)-(during-the-preceding-hour-but-not-at-the-time-of-observation)
- 23-Rain-and-snow-or-ice-pellets-not-falling-as-shower(s)-(during-the-preceding-hour-but-not-at-the-time-of-observation)
- 24-Freezing-drizzle-or-freezing-rain-not-falling-as-shower(s)-(during-the-preceding-hour-but-not-at-the-time-of-observation)
- 25-Shower(s)-of-rain-(during-the-preceding-hour-but-not-at-the-time-of-observation)
- 26-Shower(s)-of-snow-or-of-rain-and-snow-(during-the-preceding-hour-but-not-at-the-time-of-observation)

27-Shower(s)-of-hail-(Hail,-small-hail,-snow-pellets),-or-rain-and-hail-(during-the-preceding-hour-but-not-at-the-time-of-observation)

28-Fog-or-ice-fog-(during-the-preceding-hour-but-not-at-the-time-of-observation)

29-Thunderstorm-(with-or-without-precipitation)-(during-the-preceding-hour-but-not-at-the-time-of-observation)

DU:30-Slight-or-moderate-duststorm-or-sandstorm-has-decreased-during-the-preceding-hour

DU:31-Slight-or-moderate-duststorm-or-sandstorm-no-appreciable-change-during-the-preceding-hour

DU:32-Slight-or-moderate-duststorm-or-sandstorm-has-begun-or-has-increased-during-the-preceding-hour

DU:33-Severe-duststorm-or-sandstorm-has-decreased-during-the-preceding-hour

DU:34-Severe-duststorm-or-sandstorm-no-appreciable-change-during-the-preceding-hour

DU:35-Severe-duststorm-or-sandstorm-has-begun-or-has-increased-during-the-preceding-hour

DRSN:36-Slight-or-moderate-drifting-snow-generally-low-(below-eye-level)

DRSN:37-Heavy-drifting-snow-generally-low-(below-eye-level)

BLSN:38-Slight-or-moderate-blowing-snow-generally-high-(above-eye-level)

BLSN:39-Heavy-blowing-snow-generally-high-(above-eye-level)

FG:40-Fog-or-ice-fog-at-a-distance-at-the-time-of-observation,-but-not-at-the-station-during-the-preceding-hour
,the-fog-or-ice-fog-extending-to-a-level-above-that-of-the-observer

FG:41-Fog-or-ice-fog-in-patches

FG:42-Fog-or-ice-fog,-sky-visible,-has-become-thinner-during-the-preceding-hour

FG:43-Fog-or-ice-fog,-sky-invisible,-has-become-thinner-during-the-preceding-hour

FG:44-Fog-or-ice-fog,-sky-visible,-no-appreciable-change-during-the-preceding-hour

FG:45-Fog-or-ice-fog,-sky-invisible,-no-appreciable-change-during-the-preceding-hour

FG:46-Fog-or-ice-fog,-sky-visible,-has-begun-or-has-become-thicker-during-the-preceding-hour

FG:47-Fog-or-ice-fog,-sky-invisible,-has-begun-or-has-become-thicker-during-the-preceding-hour

FG:48-Fog,-depositing-rime,-sky-visible

FG:49-Fog,-depositing-rime,-sky-invisible

DZ:50-Drizzle,-not-freezing,-intermittent,-slight-at-time-of-observation

DZ:51-Drizzle,-not-freezing,-continuous,-slight-at-time-of-observation

DZ:52-Drizzle,-not-freezing,-intermittent,-moderate-at-time-of-observation

DZ:53-Drizzle,-not-freezing,-continuous,-moderate-at-time-of-observation

DZ:54-Drizzle,-not-freezing,-intermittent,-heavy-(dense)-at-time-of-observation

DZ:55-Drizzle,-not-freezing,-continuous,-heavy-(dense)-at-time-of-observation

FZDZ:56-Drizzle,-freezing,-slight

FZDZ:57-Drizzle,-freezing,-moderate-or-heavy-(dense)

DZ:58-Drizzle-and-rain,-slight

DZ:59-Drizzle-and-rain,-moderate-or-heavy

RA:60-Rain,-not-freezing,-intermittent,-slight-at-time-of-observation

RA:61-Rain,-not-freezing,-continuous,-slight-at-time-of-observation

RA:62-Rain,-not-freezing,-intermittent,-moderate-at-time-of-observation

RA:63-Rain,-not-freezing,-continuous,-moderate-at-time-of-observation

RA:64-Rain,-not-freezing,-intermittent,-heavy-at-time-of-observation

RA:65-Rain,-not-freezing,-continuous,-heavy-at-time-of-observation

FZRA:66-Rain,-freezing,-slight

FZRA:67-Rain,-freezing,-moderate-or-heavy

RA:68-Rain-or-drizzle-and-snow,-slight

RA:69-Rain-or-drizzle-and-snow,-moderate-or-heavy

SN:70-Intermittent-fall-of-snowflakes,-slight-at-time-of-observation

SN:71-Continuous-fall-of-snowflakes,-slight-at-time-of-observation

SN:72-Intermittent-fall-of-snowflakes,-moderate-at-time-of-observation

SN:73-Continuous-fall-of-snowflakes,-moderate-at-time-of-observation

SN:74-Intermittent-fall-of-snowflakes,-heavy-at-time-of-observation
 SN:75-Continuous-fall-of-snowflakes,-heavy-at-time-of-observation
 76-Diamond-dust-(with-or-without-fog)
 SG:77-Snow-grains-(with-or-without-fog)
 SN:78-Isolated-star-like-snow-crystals-(with-or-without-fog)
 PL:79-Ice-pellets
 SHRA:80-Rain-shower(s),-slight
 SHRA:81-Rain-shower(s),-moderate-or-heavy
 SHRA:82-Rain-shower(s),-violent
 SHRASN:83-Shower(s)-of-rain-and-snow-mixed,-slight
 SHRASN:84-Shower(s)-of-rain-and-snow-mixed,-moderate-or-heavy
 SHSN:85-Show-shower(s),-slight
 SHSN:86-Snow-shower(s),-moderate-or-heavy
 SH:87-Shower(s)-of-snow-pellets-or-small-hail,-with-or-without-rain-or-rain-and-snow-mixed,-slight
 SH:88-Shower(s)-of-snow-pellets-or-small-hail,-with-or-without-rain-or-rain-and-snow-mixed,-moderate-or-heavy
 SH:89-Shower(s)-of-hail-(hail,-small-hail,-snow-pellets),-with-or-without-rain-or-rain-and-snow-mixed,-not-associated-with-thunder,-slight
 SH:90-Shower(s)-of-hail-(hail,-small-hail,-snow-pellets),-with-or-without-rain-or-rain-and-snow-mixed,-not-associated-with-thunder,-moderate-or-heavy
 RA:91-Slight-rain-at-time-of-observation,-thunderstorm-during-the-preceding-hour-but-not-at-time-of-observation
 RA:92-Moderate-or-heavy-rain-at-time-of-observation,-thunderstorm-during-the-preceding-hour-but-not-at-time-of-observation
 93-Slight-snow,-or-rain-and-snow-mixed-or-hail-(Hail,-small-hail,-snow-pellets),-at-time-of-observation,-thunderstorm-during-the-preceding-hour-but-not-at-time-of-observation
 94-Moderate-or-heavy-snow,-or-rain-and-snow-mixed-or-hail-(Hail,-small-hail,-snow-pellets)-at-time-of-observation,-thunderstorm-during-the-preceding-hour-but-not-at-time-of-observation
 TS:95-Thunderstorm,-slight-or-moderate,-without-hail-(Hail,-small-hail,-snow-pellets),-but-with-rain-and/or-snow-at-time-of-observation,-thunderstorm-at-time-of-observation
 TS:96-Thunderstorm,-slight-or-moderate,-with-hail-(hail,-small-hail,-snow-pellets)-at-time-of-observation,-thunderstorm-at-time-of-observation
 TS:97-Thunderstorm,-heavy,-without-hail-(Hail,-small-hail,-snow-pellets),-but-with-rain-and/or-snow-at-time-of-observation,-thunderstorm-at-time-of-observation
 TS:98-Thunderstorm-combined-with-duststorm-or-sandstorm-at-time-of-observation,-thunderstorm-at-time-of-observation
 TS:99-Thunderstorm,-heavy,-with-hail-(Hail,-small-hail,-snow-pellets)-at-time-of-observation,-thunderstorm-at-time-of-observation

Table 5(b) pres_wx_AW1[2][3] (sourced from automated sensors)

Note: Codes 20-26 are used to report precipitation, fog, thunderstorm at the station during the preceding hour, but not at the time of observation

00-No-significant-weather-observed
 01-Clouds-generally-dissolving-or-becoming-less-developed
 02-State-of-sky-on-the-whole-unchanged-during-the-past-hr
 03-Clouds-generally-forming-or-developing-during-the-past-hr
 HZ:04-Haze,-smoke,-or-dust-in-suspension-in-the-air,-visibility-equal-to-or-greater-than-1km

FU:05

DU:07-Dust-or-sand-raised-by-wind-at-or-near-the-station-at-the-time-of-observation,-but-no-well-developed-dust-whirl(s)-or-sand-whirl(s),-and-no-duststorm-or-sandstorm-seen-or,-in-the-case-of-ships,-blowing-spray-at-the-station

BR:10-Mist

11-Diamond-dust

12-Distant-lightning

SQ:18-Squalls

20-Fog-(during-preceding-hour-but-not-at-time-of-observation)

21-Precipitation-(during-preceding-hour-but-not-at-time-of-observation)

22-Drizzle-(not-freezing)-or-snow-grains-(during-preceding-hour-but-not-at-time-of-observation)

23-Rain-(not-freezing)-or-snow-grains-(during-preceding-hour-but-not-at-time-of-observation)

24-Snow-(during-preceding-hour-but-not-at-time-of-observation)

25-Freezing-drizzle-or-freezing-rain-(during-preceding-hour-but-not-at-time-of-observation)

26-Thunderstorm-(with-or-without-precipitation)-(during-preceding-hour-but-not-at-time-of-observation)(during-preceding-hour-but-not-at-time-of-observation)

27-Blowing-or-drifting-snow-or-sand

28-Blowing-or-drifting-snow-or-sand,-visibility-equal-to-or-greater-than-1-km

29-Blowing-or-drifting-snow-or-sand,-visibility-less-than-1-km

FG:30-Fog

FG:31-Fog-or-ice-fog-in-patches

FG:32-Fog-or-ice-fog,-has-become-thinner-during-the-past-hour

FG:33-Fog-or-ice-fog,-no-appreciable-change-during-the-past-hour

FG:34-Fog-or-ice-fog,-has-begun-or-become-thicker-during-the-past-hour

FG:35-Fog,-depositing-rime

40-Precipitation

41-Precipitation,-slight-or-moderate

42-Precipitation,-heavy

43-Liquid-precipitation,-slight-or-moderate

44-Liquid-precipitation,-heavy

45-Solid-precipitation,-slight-or-moderate

46-Solid-precipitation,-heavy

47-Freezing-precipitation,-slight-or-moderate

48-Freezing-precipitation,-heavy

DZ:50-Drizzle

DZ:51-Drizzle,-not-freezing,-slight

DZ:52-Drizzle,-not-freezing,-moderate

DZ:53-Drizzle,-not-freezing,-heavy

FZDZ:54-Drizzle,-freezing,-slight

FZDZ:55-Drizzle,-freezing,-moderate

FZDZ:56-Drizzle,-freezing,-heavy

DZ:57-Drizzle-and-rain,-slight

DZ:58-Drizzle-and-rain,-moderate-or-heavy

RA:60-Rain

RA:61-Rain,-not-freezing,-slight

RA:62-Rain,-not-freezing,-moderate

RA:63-Rain,-not-freezing,-heavy

FZRA:64-Rain,-freezing,-slight

FZRA:65-Rain,-freezing,-moderate

FZRA:66-Rain,-freezing,-heavy
 RA:67-Rain-or-drizzle-and-snow,-slight
 RA:68-Rain-or-drizzle-and-snow,-moderate-or-heavy
 SN:70-Snow
 SN:71-Snow,-slight
 SN:72-Snow,-moderate
 SN:73-Snow,-heavy
 PL:74-Ice-pellets,-slight
 PL:75-Ice-pellets,-moderate
 PL:76-Ice-pellets,-heavy
 SG:77-Snow-grains
 IC:78-Ice-crystals
 80-Showers-or-intermittent-precipitation
 SHRA:81-Rain-showers-or-intermittent-rain,-slight
 SHRA:82-Rain-showers-or-intermittent-rain,-moderate
 SHRA:83-Rain-showers-or-intermittent-rain,-heavy
 SHRA:84-Rain-showers-or-intermittent-rain,-violent
 SHSN:85-Snow-showers-or-intermittent-snow,-slight
 SHSN:86-Snow-showers-or-intermittent-snow,-moderate
 SHSN:87-Snow-showers-or-intermittent-snow,-heavy
 89-Hail
 TS:90-Thunderstorm
 TS:91-Thunderstorm,-slight-or-moderate,-with-no-precipitation
 TS:92-Thunderstorm,-slight-or-moderate,-with-rain-showers-and/or-snow-showers
 TS-HAIL:93-Thunderstorm,-slight-or-moderate,-with-hail
 TS:94-Thunderstorm,-heavy,-with-no-precipitation
 TS:95-Thunderstorm,-heavy,-with-rain-showers-and/or-snow
 TS+HAIL:96-Thunderstorm,-heavy,-with-hail
 +FC:99-Tornado

Table 5(c) pres_wx_AU1[2][3] (sourced from automated ASOS/AWOS sensors)

Codes can be a sequential order/combination of several sub-codes for weather intensity, descriptor, precipitation, obscuration and other:

Intensity

1:- (light)
 3:+ (heavy)
 4:VC (vicinity-(apparent-but-not-at-point-of-observation))

Descriptor

MI = shallow
 PR = partial
 BC = patches
 DR = low-drifting
 BL = blowing
 SH = showers
 TS = thunderstorm
 FZ = freezing

Precipitation

DZ:01-Drizzle

RA:02-Rain

SN:03-Snow

SG:04-Snow-Grains

IC:05-Ice-Crystals

PL:06-Ice-Pellets

GR:07-Hail

GS:08-Small-Hail-and/or-Snow-Pellets

UP:09-Unknown-Precipitation

Obscuration

BR:1-Mist

FG:2-Fog

FU:3-Smoke

VA:4-Volcanic-Ash

DU:5-Widespread-Dust

SA:6-Sand

HZ:7-Haze

PY:8-Spray

Other weather:

PO:1-Well-developed-dust/sand-whirls

SQ:2-Squalls

FC:3-Funnel-Cloud,-Waterspout-or-Tornado

SS:4-Sandstorm

DS:5-Duststorm